Generative AI in the Software Development Lifecycle

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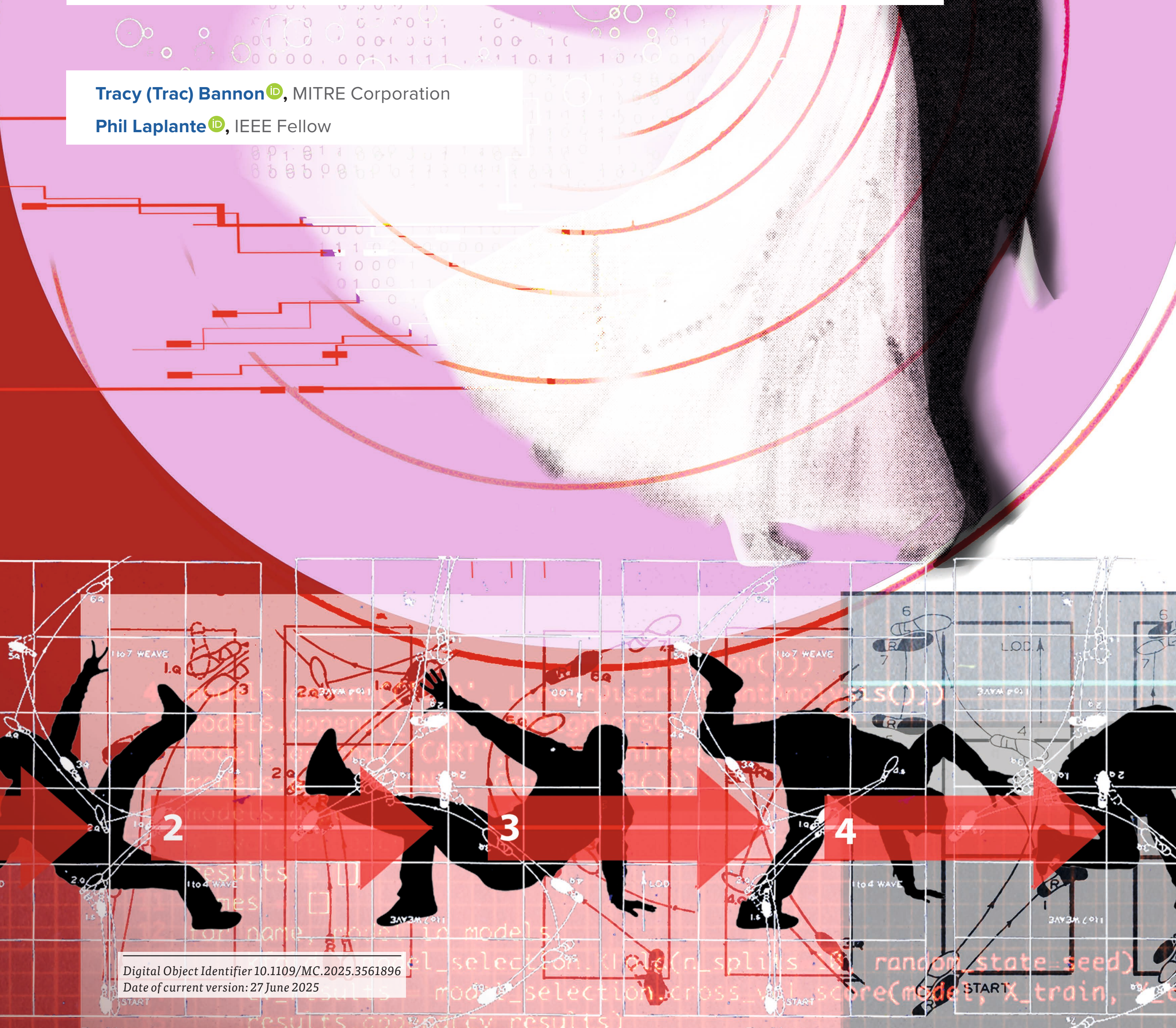

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This special issue is a thoughtful exploration of how generative AI is reshaping software engineering—highlighting the balance between automation and human judgment across the modern software development lifecycle (SDLC).

Just because we can automate something doesn't mean we should. There, we said it. Automate what's repeatable. Let humans drive what's strategic. That's the guiding principle behind our exploration of generative artificial intelligence's (AI's) role in software engineering and much of what is borne out by the papers offered by the contributors. As guest editors of this special issue, we've seen firsthand how AI is both accelerating development and raising important questions. Yes, large language models (LLMs) can assist developers by automating code generation and assisting with bug fixes, but they also risk eroding foundational skills like problem decomposition, architectural reasoning, and critical thinking.[1]

In DevSecOps and modern software practices, we love automation. It can speed up deployments, improve consistency, and help us scale. But we need to be clear about where automation shines—and where human cognition is still irreplaceable.

The temptation to automate everything is real—especially given the myriad industry tool reports. But researchers and practitioners need to cut through the hype and focus on data-driven insights. We must recognize that not all work fits into a checklist. When architects and engineers design architectures, assess risk, or troubleshoot failures, we aren't following a linear script. We are synthesizing information, weighing tradeoffs, and making decisions in complex, ever-changing environments. Yet, as DevSecOps has taught us, it's not just about speed but about doing it right. Automating too much too fast can lead to skills atrophy, collaboration breakdowns, technical debt, and even code vulnerabilities. A recent Microsoft study even suggests that the more users trust AI, the less they challenge it—raising the stakes for thoughtful integration.[2]

One technique that helps us reframe this balance is goal-directed task analysis (GDTA). Instead of breaking work into predefined steps, GDTA focuses on three elements: what goals engineers must achieve, what decisions need to be made, and what information is necessary to make those decisions effectively.[3] Applying this thinking to DevSecOps and modern software practices changes the way we think about human involvement. We must care about the value stream, remove friction, and focus on the humans in the loop today and humans on the loop in the future.

Our challenge to the community: balance innovation with intentionality, speed with sustainability, and delegation with discernment. Let's embrace the possibilities without losing the human touch that makes for great software engineering. After all, as we were reminded of a recent story of Italian mafia godfathers

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lamenting the lost skills of the younger generation—where even the basics of wielding a gun or committing a robbery are being forgotten[4]—it's a cautionary tale: When skills aren't practiced, they fade. "Leave the gun, take the cannoli"—this reminder shouldn't even need to be given, but some skills, even the most obvious, must be taught and practiced. We're certainly not comparing all software engineers to mobsters, but the point holds—let's not lose our foundational skills while embracing the new.

We hope this special issue inspires both reflection and action. Dive in and see how AI is transforming our field—strategically, thoughtfully, and with a bit of humor along the way.

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